

SANGJUN LEE

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EDUCATION

University of California, Berkeley

MAY 2027

Bachelor of Science in Electrical Engineering and Computer Science

Military Leave of Absence: Fall 2023 - Spring 2025

Relevant Coursework: Operating System and Systems Programming, Network and Internet, Database, Data Structures, Algorithms, Computer Architecture, Computer Security, Designing Information Devices and Systems, Discrete Mathematics, Probability, Multivariable Calculus, Structure and Interpretation of Computer Programs, Data Science

SKILLS

Languages: Python, Java, C, C++. Kotlin, JavaScript, TypeScript, HTML, CSS, Swift, Go, SQL, RISC-V Assembly

Frameworks & Tools: Spring Boot, Linux, React, Node.js, Express.js, FastAPI, Docker, Git, PyTorch, NumPy, Pandas, PostgreSQL, CUDA

EXPERIENCE

Google

MAY 2026 - Current

Software Engineering/Site Reliability Engineer Intern

- Developing a CLI proxy to improve SRE safety for mutating database queries by enforcing validation, guardrails, and rollback support

Sony Interactive Entertainment x Codebase

JAN 2026 - MAY 2026

Contract Software Engineer/Project Lead

- Led ML and rules-based improvements for Sony's Text Profanity Filter 2 (TPF2) pipeline, supporting online IDs, chat, bios, and user-generated content while scaling from ~2M to 70M-100M+ daily requests and up to 200M peak requests
- Implemented rules-based algorithms in Java to detect profane emoji combinations, typo-tolerant profanity, and obfuscation bypasses, helping expand TPF2 from 4 to 37 production locales and fully migrate from TPF1 legacy filtering pipeline
- Built a language-agnostic data preprocessing pipeline and fine-tuned Arabic/Japanese models for context-aware profanity detection

Google x Codebase

SEP 2025 - MAR 2026

Contract Software Engineer

- Developed and enhanced open-source tools for the AlloyDB Omni Operator, enabling Kubernetes users to improve diagnostics, performance analysis, and resilience testing of on-premise database deployments
- Built observability dashboards (Grafana/Prometheus/KSM) visualizing cluster health, backup workflows, and operator features
- Collaborated in a cross-functional team to contribute production-ready Golang and Kubernetes API utilities for database monitoring

Samsung

JUNE 2025 - AUG 2025

Software Engineer Intern

South Korea

- Engineered a novel GUI automation agent to overcome the API and security limitations of traditional AI agents, successfully automating complex workflows of over 15 actions and cutting manual task completion time by approximately 80%
- Built a FastAPI and React system integrating multiple MLLM models with PyAutoGUI to translate user prompts into on-screen actions
- Boosted MLLM accuracy (Llama 4, UGround) by 30% through systematic prompt engineering and testing of over 20 distinct variations

Republic of Korea Air Force Headquarters

OCT 2023 - MAY 2025

Backend Developer

South Korea

- Served as a backend developer for an automated runway crack detection system, reducing inspection and labeling time by over 95%
- Implemented a Spring Boot backend with Oracle Database, managing 100,000+ runway slabs and permissions for 5,000+ users
- Led the full migration of the backend system from Java to Kotlin to boost long-term maintainability and enhance code safety
- Deployed an inference pipeline that uses YOLOv8 & OpenCV to label 4000+ images per hour, increasing detection accuracy by 35%

Machine Learning Engineer

South Korea

- Developed the FastAPI backend for a real-time transcription service, leveraging multithreading and concurrency to handle simultaneous model requests efficiently, reducing manual transcription time from weeks to minutes for 200+ daily users
- Increased transcription accuracy by 50% by training and fine-tuning STT and diarization models using PyTorch on large-scale datasets
- Boosted deployment time by 200%, using Docker for containerization and Kubernetes for orchestration and automated deployments
- Awarded the Prime Minister Award as part of the team that substantially reduced soldiers' administrative workload through AI

TEACHING & LEADERSHIP

Data 8 Course Staff

AUG 2025 - Current

Course Director, Grading Lead, Undergraduate Course Instructor

Berkeley, CA

- Served as a Course Director, leading staff operations and coordinating with professors on logistics, exams, and course content
- Rebuilt the grading codebase to automate final grade calculations, section analysis, and grade reports for 1500+ students

CS 61C Course Staff

JUNE 2023 - AUG 2023

Undergraduate Student Instructor

Berkeley, CA

- Instructed for a computer architecture class of 200+ students, leading to a 90% increase in student participation in discussion sections
- Created weekly discussion worksheets on core concepts such as C, RISC-V, hardware, and optimization (SIMD, OpenMP, OpenMPI)